UNIT-II

VECTOR INTEGRATION

Topic Learning Objectives:

Upon Completion of this unit, students will be able to:

- Understand the fundamentals of the integration of vector point function.
- Solve line, surface and volume integrals.
- Apply Green's Theorem, Stokes' Theorem and Gauss' Theorem in solving engineering problems.
- Estimate and apply the concepts of solenoidal and irrotational fields to calculate integrals of vector functions.

Line Integral: Any integral which is to be evaluated along a curve is called line integral.

If $\vec{F}(x,y,z)$ is a vector point function and C is any curve then $\int_C \vec{F}.d\vec{r}$ is called the vector line integral. (Tangential line integral or line integral)

NOTE:

1. C is a called path of integration.
2. If $\vec{F} = f_1\hat{\imath} + f_2\hat{\jmath} + f_3\hat{k}$ then $\int_C \vec{F}.d\vec{r} = \int_C f_1dx + f_2dy + f_3dz$.
3. When C is a simple closed curve, line integral is denoted by $\oint_C \vec{F}.d\vec{r}$ (means the line integral of $\vec{F}$ taken once around C in the anticlock wise direction).
4. If $\vec{F}$ represents force acting on a particle then the line integral $\int_C \vec{F}.d\vec{r}$ represents work done by a force $\vec{F}$.
5. If $\vec{F}$ represents the velocity of a fluid then $\int_C \vec{F}.d\vec{r}$ represents circulation of $\vec{F}$ around C.
6. Condition for $\vec{F}$ to be conservative is $\nabla \times \vec{F} = 0$.
7. If $curl\,\vec{F} = 0$ then $\int_C \vec{F}.d\vec{r}$ is independent of path.

Problem 1. If $\vec{F} = (5xy - 6x^2)\hat{\imath} + (2y - 4x)\hat{\jmath}$. Evaluate $\int_C \vec{F}.d\vec{r}$ along $y = x^3$ in XY −plane from $(1,1)$ to $(2,8)$.

Solution: Given

$\vec{F} = (5xy - 6x^2)\hat{\imath} - (2y - 4x)\hat{\jmath}$

$\vec{r} = x\,\hat{\imath} + y\hat{\jmath} \Rightarrow \overrightarrow{dr} = dx\,\hat{\imath} + dy\hat{\jmath}$

$y = x^3 \Rightarrow dy = 3x^2dx$ and x: 1 to 2

DEPARTMENT OF MATHEMATICS, RVCE

Consider

$\vec{F}.d\vec{r} = (5x\,x^3 - 6x^2)dx + (2x^3 - 4x)3x^2dx$

$$\int_C \vec{F}.d\vec{r} = \int_1^2 (5x^4 - 6x^2 + 6x^5 - 12x^3)dx = [x^5 - 2x^3 + x^6 - 3x^4]_1^2 = 35.$$

Problem 2. Find the work done in moving a particle in the force field $\vec{F} = 3x^2\hat{\imath} + (2xz - y)\hat{\jmath} + z\hat{k}$ along

a) The straight line $(0,0,0)$ to $(2,1,3)$.
b) The curve $x = 2t^2, y = t, z = 4t^2 - t$ from $t = 0$ to $t = 1$.
c) The curve defined by $x^2 = 4y,\ 3x^3 = 8z$ from $x = 0$ to 2.

Solution: Work done $= \int_C \vec{F}.d\vec{r} = \int_C 3x^2dx + (2xz - y)dy + zdz.$ ---------- (i)

a) C is a straight line joining $(0,0,0)$ and $(2,1,3)$.

The equation of the line is given by $\frac{x}{2} = \frac{y}{1} = \frac{z}{3} = t$

We have $x = 2t \Rightarrow dx = 2dt,\qquad y = t \Rightarrow dy = dt,\qquad z = 3t \Rightarrow dz = 3dt$

and $t = 0$ to 1 $[\because t = y,\ y = 0\ to\ 1]$

then equation (i)

$$\Rightarrow \int_C \vec{F}.d\vec{r} = \int_0^1 \{(3(2t)^2)(2dt) + (2(2t)(3t) - t)dt + (3t)3dt\}$$

$$= \int_0^1 (36t^2 + 8t)\,dt = \left[36\frac{t^3}{3} + 8\frac{t^2}{2}\right]\Bigg|_0^1 = 16.$$

b) Given curve $x = 2t^2 \Rightarrow dx = 4\,tdt,\qquad y = t \Rightarrow dy = dt,\ \ z = 4t^2 - t \Rightarrow dz = (8t - 1)dt$

then (i) becomes

$$\int_C \vec{F}.d\vec{r} = \int_0^1 \{(3(2t^2)^2)(4tdt) + (2(2t^2)(4t^2 - t) - t)dt + (4t^2 - t)(8t - 1)dt\}$$

$$= \int_0^1 (48t^5 + 16t^4 + 28t^3 - 12t^2)\,dt = \frac{71}{5}$$

c) Given curve $x^2 = 4y \Rightarrow y = \frac{x^2}{4} \Rightarrow dy = \frac{x}{2}dx,\ 3x^3 = 8z \Rightarrow z = \frac{3x^3}{8} \Rightarrow dz = \frac{9}{8}x^2dx$ and x: 0 to 2 then (i) becomes

$$\int_C \vec{F}.d\vec{r} = \int_0^2 \left\{3x^2dx + \left(2x\left(\frac{3}{8}x^3\right) - \frac{x^2}{4}\right)\frac{x}{2}dx + \frac{3}{8}x^3\frac{9}{8}x^2\,dx\right\}$$

DEPARTMENT OF MATHEMATICS, RVCE

$$= \int_0^2 (3x^2 + \frac{3}{8}x^5 - \frac{x^3}{8} + \frac{27}{64}x^5)\, dx = 16.$$

Exercise:

1. If $\vec{F} = x^2\hat{\imath} + xy\hat{\jmath}$, evaluate $\int_C \vec{F}.d\vec{r}$ from $(0,0)$ to $(1,1)$

(a) along the line $y = x$ Ans: $\frac{2}{3}$

(b) along the parabola $y = \sqrt{x}$ Ans: $\frac{7}{12}$.

2. Find the total work done by the force represented by $\vec{F} = 3xy\hat{\imath} - y\hat{\jmath} + 2zx\hat{k}$ in moving a particle round the circle $x^2 + y^2 = 4, x = 2\cos\theta, y = 2\sin\theta$ & $z = 0, 0 \le \theta \le 2\pi$.

3. Find the circulation of $\vec{F}$ around the curve C, where C is the rectangle whose vertices are given by $(0,0), (1,0), \left(1, \frac{\pi}{2}\right)$ & $\left(0, \frac{\pi}{2}\right)$ and $\vec{F} = e^x \sin y\, \hat{\imath} + e^x \cos y\, \hat{\jmath}$.

4. If $\vec{F} = (2x + y^2)\hat{\imath} + (3y - 4x)\hat{\jmath}$ evaluate $\oint_C \vec{F} \cdot \overrightarrow{dr}$ around a triangle ABC in the xy −plane with $A(0,0)$ $B(2,0)$ and $C(2,1)$,

(a) In the counter clockwise direction. Ans: $-\frac{14}{3}$

(b) What is the value in the opposite direction? Ans: $\frac{14}{3}$

5. Evaluate the line integral $\int_C (x^2 + xy)dx + (x^2 + y^2)dy$, where $C: square: x = \pm 1,\ y = \pm 1$. Ans:0

NOTE: If circulation is "0" then $\int \vec{F}.d\vec{r}$ is irrotational.

GREEN'S THEOREM

Green's theorem in the plane transforms a line integral to a double integral in a plane.

Statement:

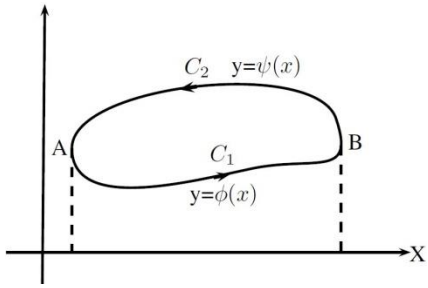
If R is a closed region in XY-plane, bounded by a simply closed curve C and if $P(x,y)$ and $Q(x,y)$, $\frac{\partial}{\partial x}Q(x,y)$, $\frac{\partial}{\partial y}P(x,y)$ be continuous functions at every point in R, then

$$\oint_C P\, dx + Q\, dy = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\, dy$$

.

Proof: Suppose that C is a simply closed curve with the property that any line parallel to either axis meets the curve in at most two points.

DEPARTMENT OF MATHEMATICS, RVCE



Consider

$$\iint_R \left(-\frac{\partial P}{\partial y}\right) dx\, dy = \int_{x=a}^{b} \int_{y=\phi(x)}^{\psi(x)} \left(-\frac{\partial P}{\partial y}\right) dy\, dx = \int_{x=a}^{b} -P(x,y)\Big|_{\phi(x)}^{\psi(x)} dx$$

$$= \int_{x=a}^{b} \left[-P\big(x,\psi(x)\big) + P\big(x,\phi(x)\big)\right] dx$$

$$= \int_{b}^{a} P\big(x,\psi(x)\big)\, dx + \int_{a}^{b} P\big(x,\phi(x)\big)\, dx$$

$$\Rightarrow \int_{c_1} P(x,y)dx + \int_{c_2} P(x,y)dx = \int_{c} P(x,y)dx$$

Similarly,

$$\iint_R \left(\frac{\partial Q}{\partial x}\right) dx\, dy = \int_{c} Q(x,y)dx$$

$$\therefore \int_{c} P(x,y)dx + Q(x,y)dy = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\, dy.$$

Problem 1. Verify Green's theorem in the plane for $\oint_C \{(x^2+y)dx - xy^2dy\}$ taken around the boundary of the rectangle whose vertices are $(0,\ 0), (a,0), (a,b)$ and $(0,b)$.

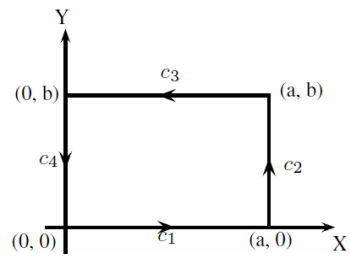
Solution: We have to verify

$$\oint_C P(x,y)dx + Q(x,y)dy = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\, dy$$

Consider

$$\oint_C Pdx + Qdy = \int_{c_1} Pdx + Qdy + \int_{c_2} Pdx + Qdy + \int_{c_3} Pdx + Qdy + \int_{c_4} Pdx + Qdy$$

DEPARTMENT OF MATHEMATICS, RVCE



$$\oint_C \{(x^2+y)dx - xy^2dy\} = \oint_C Pdx + Qdy$$

Along $C_1: y = 0 \Rightarrow dy = 0 \; and \; x: 0 \; to \; a$

$$\int_{c_1} \{(x^2+y)dx - xy^2dy\} = \int_0^a x^2dx = \left.\frac{x^3}{3}\right|_0^a = \frac{a^3}{3}.$$

Along $C_2: x = a \Rightarrow dx = 0 \; and \; y: 0 \; to \; b$

$$\int_{c_2} \{(x^2+y)dx - xy^2dy\} = \int_0^b -ay^2dy = -\left.\frac{ay^3}{3}\right|_0^b = -\frac{ab^3}{3}.$$

Along $C_3: y = b \Rightarrow dy = 0 \; and \; x: a \; to \; 0$

$$\int_{c_3} \{(x^2+y)dx - xy^2dy\} = \int_a^0 (x^2+b)\,dx = \left.\frac{x^3}{3} + bx\right|_a^0 = -\frac{a^3}{3} - ba.$$

Along $C_4: x = 0 \Rightarrow dx = 0 \; and \; y: b \; to \; 0$

$$\int_{c_3} \{(x^2+y)dx - xy^2dy\} = \int_b^0 0\,dy = 0$$

$$\therefore \oint_C \{(x^2+y)dx - xy^2dy\} = \oint_C Pdx + Qdy$$

$$= \frac{a^3}{3} - \frac{ab^3}{3} - \frac{a^3}{3} - ba + 0 = -ab\left(1 + \frac{b^2}{3}\right) \qquad \dots\dots(1)$$

Next consider,

$$\iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy = \int_{x=0}^{a}\int_{y=0}^{b} (-y^2 - 1)\,dy\,dx = -\int_{x=0}^{a}\left[\frac{y^3}{3} + y\right]_0^b dx = -\int_0^a \left(\frac{b^2}{3} + b\right) dx$$

$$= -ab\left(1 + \frac{b^2}{3}\right) \qquad \dots\dots(2)$$

DEPARTMENT OF MATHEMATICS, RVCE

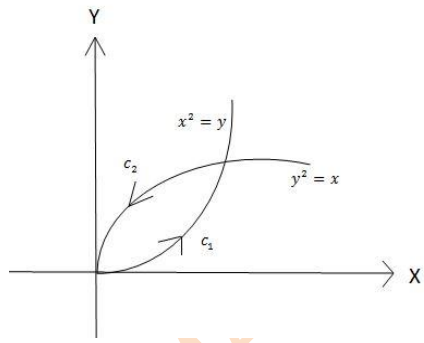
From (1) and (2), Green's theorem is verified.

Problem 2. Verify Green's theorem in the plane for $\int_c \{(x-y)dx + (x+y)dy\}$ taken around the boundary of the finite area in the positive quadrant included between $y = x^2$ & $x = y^2$.

Solution: We have to verify

$$\int_c P(x,y)dx + Q(x,y)dy = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\, dy$$

$\int_C P\, dx + Q\, dy = \int_{C_1} P\, dx + Q\, dy + \int_{C_2} P\, dx + Q\, dy = \int_{C_1} \{(x-y)dx + (x+y)dy\} + \int_{C_2} \{(x-y)dx + (x+y)dy\}$



Along C_1: $y = x^2 \Rightarrow dy = 2xdx$ and x: 0 to 1

$$\int_{C_1} \{(x-y)dx + (x+y)dy\} = \int_0^1 \{(x-x^2)dx + (x+x^2)2xdx\}$$

$$= \int_0^1 (2x^3 + x^2 + x)dx = \left[2\frac{x^4}{4} + \frac{x^3}{3} + \frac{x^2}{2}\right]_0^1 = \frac{4}{3}.$$

Along C_2: x= $y^2 \Rightarrow dx = 2ydy$ and y: 1 to 0

$$\int_{C_1} \{(x-y)dx + (x+y)dy\} = \int_1^0 \{(y^2-y)2ydy + (y^2+y)dy\}$$

$$= \int_1^0 (2y^3 - y^2 + y)dy = \left[2\frac{y^4}{4} - \frac{y^3}{3} + \frac{y^2}{2}\right]_1^0 = -\frac{2}{3}.$$

$$\therefore \int_C P\, dx + Q\, dy = \frac{4}{3} - \frac{2}{3} = \frac{2}{3} \qquad \ldots\ldots\ldots\ldots(1)$$

Now $P = x - y \Rightarrow \frac{\partial P}{\partial y} = -1$ and $Q = x + y \Rightarrow \frac{\partial Q}{\partial x} = 1$

$$\iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\, dy = \int_{y=0}^{1} \int_{x=y^2}^{\sqrt{y}} (1+1)\, dx\, dy = \int_{y=0}^{1} \left(\int_{x=y^2}^{\sqrt{y}} 2\, dx \right) dy$$

$$= \int_{y=0}^{1} (2x)_{y^2}^{\sqrt{y}} dy = 2\int_{y=0}^{1} \left(\sqrt{y} - y^2\right) dy$$

$$= 2\left(\frac{y^{\frac{3}{2}}}{\frac{3}{2}} - \frac{y^3}{3}\right)_0^1 = \frac{2}{3} \qquad \ldots\ldots\ldots\ldots(2)$$

From (1) and (2), Green's theorem is verified.

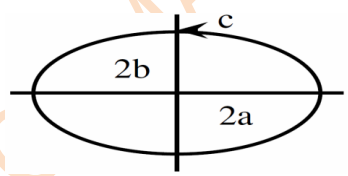
Problem 3. Show that area enclosed by a simple closed curve C is given by $\frac{1}{2}\oint\{xdy - ydx\}$. Using this, find the area bounded by the ellipse with axes $2a$ and $2b$.

Solution: we have

$$\oint_C Pdx + Qdy = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\, dy$$

$P = -y \Rightarrow \frac{\partial P}{\partial y} = -1$ and $Q = x \Rightarrow \frac{\partial Q}{\partial x} = 1$

$$\frac{1}{2}\oint_C xdy - ydx = \frac{1}{2}\iint_R (1+1)dx\, dy$$



To find the area of the ellipse, the equation is $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

In parametric form

$x = a\cos\theta, \qquad y = b\sin\theta.$

$\Rightarrow dx = -a\sin\theta, \qquad dy = b\cos\theta$

$$Area = \frac{1}{2}\oint_C xdy - ydx = \frac{1}{2}\oint_C \{a\cos\theta\, b\cos\theta - b\sin\theta\,(-a\sin\theta)\}\, d\theta$$

$$= \frac{1}{2}ab\int_0^{2\pi} (\cos^2\theta + \sin^2\theta)d\theta = \frac{1}{2}ab[2\pi] = \pi ab.$$

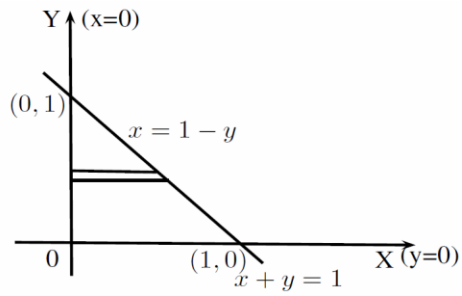
Problem 4. Using Green's theorem in the plane, evaluate $\int_C \{(2x^2 - y^2)dx + (x^2 + y^2)dy\}$, C is the boundary of the region bounded by $x = 0, y = 0, x + y = 1$.

Solution: we have

DEPARTMENT OF MATHEMATICS, RVCE

$$\oint_C Pdx + Qdy = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\, dy$$

$P = 2x^2 - y^2 \Rightarrow \frac{\partial P}{\partial y} = -2y$ and $Q = x^2 + y^2 \Rightarrow \frac{\partial Q}{\partial x} = 2x$



$$\iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\, dy = \int_{y=0}^{1}\int_{x=0}^{1-y} (2x + 2y)dxdy = 2\int_0^1 \left[\frac{x^2}{2} + xy\right]_0^{1-y} dy$$

$$= 2\int_0^1 \left(\frac{(1-y)^2}{2} + (1-y)y\right) dy$$

$$= 2\int_0^1 \left(\frac{1}{2}(1 + y^2 - 2y) + (y - y^2)\right) dy$$

$$= 2\left[\frac{1}{2}\left(y + \frac{y^3}{3} - 2\frac{y^2}{2}\right) + \left(\frac{y^2}{2} - \frac{y^3}{3}\right)\right]_0^1 = \frac{2}{3}.$$

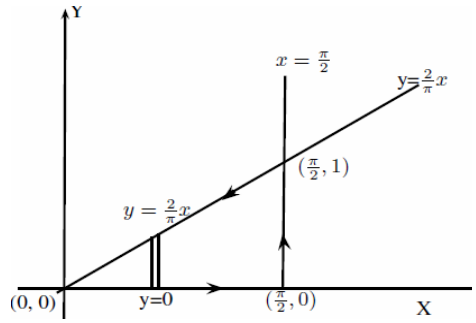
Problem 5. Apply Green's theorem to evaluate $\int_c (y - \sin x)dx + \cos x\, dy$, where C is the triangle enclosed by the lines $y = 0, x = \frac{\pi}{2}, and\ y = \frac{2}{\pi}x$.

Solution: we have

$$\oint_C Pdx + Qdy = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\, dy$$

$P = y - \sin x \Rightarrow \frac{\partial P}{\partial y} = 1$ and $Q = \cos x \Rightarrow \frac{\partial Q}{\partial x} = -\sin x$

DEPARTMENT OF MATHEMATICS, RVCE



$$\int_C (y - \sin x)dx + \cos x\, dy = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\, dy$$

$$= \int_{x=0}^{\frac{\pi}{2}} \int_{y=0}^{\frac{2x}{\pi}} (-\sin x - 1)dy\, dx = -\int_0^{\frac{\pi}{2}} (\sin x + 1)[y]_0^{\frac{2x}{\pi}}\, dx$$

$$= -\frac{2}{\pi}\int_0^{\frac{\pi}{2}} (x\ \sin x + x)\, dx$$

$$= -\frac{2}{\pi}\left[-x\cos x + \sin x + \frac{x^2}{2}\right]_0^{\frac{\pi}{2}} = -\left(\frac{2}{\pi} + \frac{\pi}{4}\right).$$

Exercise:

1. Verify Green's theorem for $\int_C (e^{-x}\sin y)dx + (e^{-x}\cos y)dy$, where C is the rectangle, whose vertices are $(0,0), (\pi, 0), \left(\pi, \frac{\pi}{2}\right)$ *and* $\left(0, \frac{\pi}{2}\right)$. Ans: $[2(e^{-\pi} - 1)]$

2. Using Green's theorem, evaluate $\oint_C\ x^{-1}e^y dx + (e^y \ln x + 2x\ )dy$, where C is the bounded by $y = 2, y = x^4 + 1$. *Ans*: $\frac{16}{5}$

3. Using Green's theorem, evaluate $\oint_C\ (x^2 - \cosh y\, dx + (y + sinx)dy)$ where C is the boundary of the rectangle $0 \le x \le \pi,\ 0 \le y \le 1$. *Ans*: $\pi(\cosh 1 - 1)$

DEPARTMENT OF MATHEMATICS, RVCE

Surface Integral

Any integral which is to be evaluated over a surface is called surface integral.

Physical interpretation: The surface integral of a vector function $\vec{F}$ express the normal flux through a surface.

Note: If $\vec{F}$ represents velocity vector of a fluid, the surface integral represents the rate of flow of fluid through the surface.

1. The surface integral of a vector point function $\vec{F}$ over a surface S is defined as the integral of normal component of $\vec{F}$ taken over the surface S.
2. If $\vec{F}$ represents the velocity of a fluid $\oiint_S \vec{F}.\hat{n}\, ds$ gives the flux across the surface S.
3. If the flux of $\vec{F}$ across every closed surface S in a region R is zero. Then $\vec{F}$ is a solenoidal vector point function in the region R.
4. If $\vec{F}$ represents gravitational force, electric force or magnetic force in each case $\iint_S \vec{F}.\hat{n}\, ds$ gives corresponding flux.

Working Rule:

1. For the given surface $\emptyset$, find $\hat{n} = \frac{\nabla\emptyset}{|\nabla\emptyset|}$, $\hat{n}$ is outward unit normal vector to the surface.
2. Find $\vec{F}.\hat{n}$
3. If the projection of S is taken in YZ −plane, then $ds = \frac{dy\, dz}{|\hat{n}.\hat{\imath}|}$, where $\hat{\imath}$ is the unit vector along $x - axis$.
4. If the projection of S is taken in XY −plane, then $ds = \frac{dx\, dy}{|\hat{n}.\hat{k}|}$, where $\hat{k}$ is the unit vector along $z - axis$.
5. If the projection of S is taken in XZ −plane, then $ds = \frac{dx\, dz}{|\hat{n}.\hat{\jmath}|}$, where $\hat{\jmath}$ is the unit vector along $y - axis$.

NOTE: To evaluate any surface integral, it is convenient to evaluate the double integral of its projection on $xy, yz, or\ zx$ plane.

$$\iint_S \vec{F}.\hat{n}\, ds = \iint_S \vec{F}.\overrightarrow{ds} = \iint_R \vec{F}.\hat{n}\, \frac{dx\, dy}{|\hat{n}.\hat{k}|}$$

where R is the projection of S in $XY - plane$.

DEPARTMENT OF MATHEMATICS, RVCE

Problems 1. Evaluate $\iint_S \vec{F}.\hat{n}\, ds$ where $\vec{F} = 18z\hat{\imath} - 12\hat{\jmath} + 3y\hat{k}$ and S is the part of the $2x + 3y + 6z = 12$, located in first octant $(x = 0, y = 0, z = 0)$.

Solution: Given $2x + 3y + 6z = 12 \Rightarrow \frac{x}{6} + \frac{y}{4} + \frac{z}{2} = 1$

Let $\emptyset = 2x + 3y + 6z - 12$

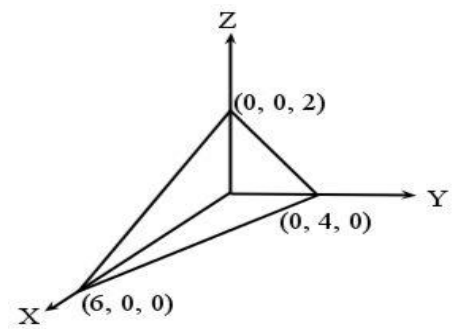
then $\nabla\emptyset = \frac{\partial\emptyset}{\partial x}\hat{\imath} + \frac{\partial\emptyset}{\partial y}\hat{\jmath} + \frac{\partial\emptyset}{\partial z}\hat{k} = 2\hat{\imath} + 3\hat{\jmath} + 6\hat{k}$

$$\hat{n} = \frac{\nabla\emptyset}{|\nabla\emptyset|} = \frac{2\hat{\imath} + 3\hat{\jmath} + 6\hat{k}}{\sqrt{2^2 + 3^2 + 6^2}} = \frac{2\hat{\imath} + 3\hat{\jmath} + 6\hat{k}}{7}$$

$$\vec{F}.\hat{n} = \left(18z\hat{\imath} - 12\hat{\jmath} + 3y\hat{k}\right).\left(\frac{2\hat{\imath} + 3\hat{\jmath} + 6\hat{k}}{7}\right) = \frac{36z - 36 + 18y}{7}$$

Projecting on to any plane (i.e, xy, yz or zx)

Projecting on to plane $xy - plane$



$2x + 3y = 12;\quad x: 0\ to\ 6\ ;\quad y: 0\ to\ \frac{12-2x}{3}$ and $|\hat{n}.\hat{k}| = \frac{6}{7}$

$$\iint_S \vec{F}.\hat{n}\, ds = \iint_R \vec{F}.\hat{n}\, \frac{dx\, dy}{|\hat{n}.\hat{k}|} = \int_{x=0}^{6}\int_{y=0}^{\frac{12-2x}{3}} \frac{36z - 36 + 18y}{7}\, \frac{dy\, dx}{\frac{6}{7}}$$

$$= \frac{1}{6}\int_{x=0}^{6}\int_{y=0}^{\frac{12-2x}{3}} \left(36\, \frac{12 - 2x - 3y}{6} - 36 + 18y\right) dy\, dx$$

$$= \frac{1}{6}\int_{x=0}^{6}\int_{y=0}^{\frac{12-2x}{3}} (36 - 12x)\, dy\, dx = \frac{1}{6}\int_{x=0}^{6} (36\, y - 12\, x\, y)\Big|_0^{\frac{12-2x}{3}} dx$$

$$= 2\int_{x=0}^{6} (3 - x\,)y\Big|_0^{\frac{12-2x}{3}} dx = 2\int_0^6 (3 - x\,)\left(\frac{12 - 2x}{3} - 0\right) dx$$

$$= 2\int_0^6 (3-x)\,\frac{12-2x}{3}\,dx = 24.$$

Problem 2. Evaluate $\iint_S \vec{F}.\hat{n}\,ds$ where $\vec{F} = y\hat{\imath} + 2x\hat{\jmath} - z\hat{k}$ and S is the surface of the plane $2x + y = 6$ included in the I octant cut by $z = 4$.

Solution: Let $\emptyset = 2x + y - 6$

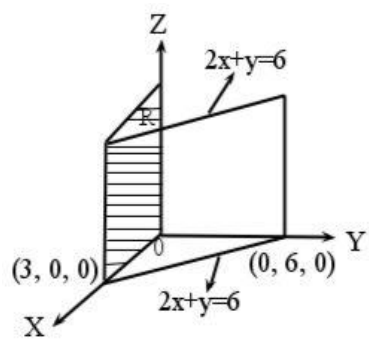
then $\nabla\emptyset = \frac{\partial\emptyset}{\partial x}\hat{\imath} + \frac{\partial\emptyset}{\partial y}\hat{\jmath} + \frac{\partial\emptyset}{\partial z}\hat{k} = 2\hat{\imath} + \hat{\jmath}$

$$\hat{n} = \frac{\nabla\emptyset}{|\nabla\emptyset|} = \frac{2\hat{\imath}+\hat{\jmath}}{\sqrt{4+1}} = \frac{2\hat{\imath}+\hat{\jmath}}{\sqrt{5}}$$

$$\vec{F}.\hat{n} = \left(y\hat{\imath} + 2x\hat{\jmath} - z\hat{k}\right).\left(\frac{2\hat{\imath}+\hat{\jmath}}{\sqrt{5}}\right) = \frac{2y+2x}{\sqrt{5}}$$

Projecting on to $xz - plane$, we get

$|\hat{n}.\hat{\jmath}| = \frac{1}{\sqrt{5}}$; $x: 0\ to\ 3$ and $z: 0\ to\ 4$



Now consider

$$\iint_S \vec{F}.\hat{n}\,ds = \iint_R \vec{F}.\hat{n}\,\frac{dx\,dz}{|\hat{n}.\hat{\jmath}|}$$

$$= \int_{x=0}^{3}\int_{z=0}^{4} \frac{2y+2x}{\sqrt{5}}\,\frac{dz\,dx}{\frac{1}{\sqrt{5}}} = \int_{x=0}^{3}\int_{z=0}^{4} (2(6-2x)+2x)\,dz\,dx$$

$$= \int_{x=0}^{3}\int_{z=0}^{4} (12-2x)\,dz\,dx = \int_{x=0}^{3}(12-2x)\,dx \int_{z=0}^{4} 1\,dz = 108.$$

Problem 3. Evaluate $\iint_S \vec{F}.\hat{n}\,ds$ where $\vec{F} = z\hat{\imath} + x\hat{\jmath} + 3y^2z\hat{k}$ where S is the surface of the cylinder $x^2 + y^2 = 16$ included in the first octant between $z = 0$ and $z = 5$.

Solution: Given $x^2 + y^2 = 16$ is a right circular cylinder with base circle as $x^2 + y^2 = 16,\ z = 0$ and generates parallel to $z - axis$.

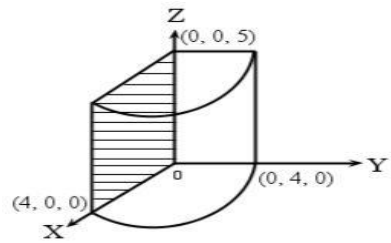
Let $\emptyset = x^2 + y^2 - 16$

DEPARTMENT OF MATHEMATICS, RVCE

then $\nabla\emptyset = \frac{\partial\emptyset}{\partial x}\hat{\imath} + \frac{\partial\emptyset}{\partial y}\hat{\jmath} + \frac{\partial\emptyset}{\partial z}\hat{k} = 2x\hat{\imath} + 2y\hat{\jmath}$

$$\hat{n} = \frac{\nabla\emptyset}{|\nabla\emptyset|} = \frac{2x\hat{\imath} + 2y\hat{\jmath}}{\sqrt{4x^2 + 4y^2}} = \frac{2(x\hat{\imath} + y\hat{\jmath})}{2\sqrt{(x^2 + y^2)}} = \frac{x\hat{\imath} + y\hat{\jmath}}{\sqrt{16}} = \frac{x\hat{\imath} + y\hat{\jmath}}{4}$$

$$\vec{F}.\hat{n} = (z\hat{\imath} + x\hat{\jmath} + 3y^2z\hat{k}).\left(\frac{x\hat{\imath} + y\hat{\jmath}}{4}\right) = \frac{\text{xz} + \text{xy}}{4}$$



Projecting on to plane $xz - plane$

$$\iint_S \vec{F}.\hat{n}\, ds = \iint_R \vec{F}.\hat{n}\, \frac{dx\, dz}{|\hat{n}.\hat{\jmath}|} = \int_{z=0}^{5}\int_{x=0}^{4} \frac{\text{xz} + \text{xy}}{4}\, \frac{dx\, dz}{\frac{y}{4}} \quad \because |\hat{n}.\hat{k}| = \frac{y}{4}$$

$$= \int_{z=0}^{5}\int_{x=0}^{4} \left(\frac{\text{xz}}{y} + x\right) dx\, dz = \int_{z=0}^{5}\int_{x=0}^{4} \left(\frac{\text{xz}}{\sqrt{16 - x^2}} + x\right) dx\, dz$$

$$= \int_{z=0}^{5} z\, dz \int_{x=0}^{4} \frac{\text{x}}{\sqrt{16 - x^2}}\, dx + \int_{x=0}^{4} x\, dx \int_{z=0}^{5} 1\, dz = 90.$$

Exercise:

1. Find the surface integral over the parallelepiped $x = 0, y = 0, x = 1, y = 2, z = 3$ when $\vec{A} = 2xy\hat{\imath} + yz^2\hat{\jmath} + xz\hat{k}$ $\quad Ans: 33.$

2. If S is the surface of the sphere $x^2 + y^2 + z^2 = d^2$ and $\vec{A} = ax\hat{\imath} + by\hat{\jmath} + cz\hat{k}$, evaluate $\iint_S \vec{A} \cdot \hat{n}\, ds.$ $\quad Ans: \frac{2\pi d^3}{3}(a + b + c)$

3. If $\vec{F} = 2y\hat{\imath} - 3\hat{\jmath} + x^2\hat{k}$ and S is the surface of the parabolic cylinder $y^2 = 8x$ in the first octant bounded by the planes $y = 4$ and $z = 6$, show that $\iint_S \vec{F} \cdot \hat{n}\, ds = 132.$

DEPARTMENT OF MATHEMATICS, RVCE

Gauss divergence theorem

(Relation between surface and volume integrals)

Statement: If V is the volume bounded by a closed surface S and $\vec{F}$ is a vector point function having continuous derivatives, then

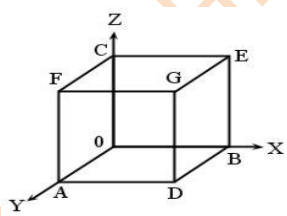
$$\iint_S \vec{F}.\hat{n}\, ds = \iiint_V \nabla.\vec{F}\, dV,$$

where $\hat{n}$ is the unit normal drawn to S. ($\hat{n} \rightarrow$ outward unit normal i.e, normal vector away from the surface)

Problem 1. Verify Gauss divergence theorem for $\vec{F} = (x^3 - yz)\hat{\imath} - 2x^2y\hat{\jmath} + z\hat{k}$ taken over the suface of the cube bounded by the planes $x = y = z = 2$ and the coordinate planes.

Solution: We have to verfy that

$$\iint_S \vec{F}.\hat{n}\, ds = \iiint_V \nabla.\vec{F}\, dV$$



$$\iint_S \vec{F}.\hat{n}\, ds = \int_{S_1} \vec{F}.\hat{n}\, ds + \int_{S_2} \vec{F}.\hat{n}\, ds + \int_{S_3} \vec{F}.\hat{n}\, ds + \int_{S_4} \vec{F}.\hat{n}\, ds + \int_{S_5} \vec{F}.\hat{n}\, ds + \int_{S_6} \vec{F}.\hat{n}\, ds$$

where $S_1,\ S_2,\ S_3,\ S_4,\ S_5,\ S_6$ are the six faces of the cube.

$$\therefore \iint_S \vec{F}.\hat{n}\, ds = \iint_S \left((x^3 - yz)\hat{\imath} - 2x^2y\hat{\jmath} + z\hat{k}\right).\hat{n}\, ds$$

For S_1 $(DBEG)$ which is parallel to $yz - plane$ its equation is $x = 2, \hat{n} = i$ & $ds = dydz$.

Here $\hat{n} = i,\quad (x^3 - yz)i.i = x^3 - yz$ (remaining are zero).

$$\iint_{S_1} \left((x^3 - yz)\hat{\imath} - 2x^2y\hat{\jmath} + z\hat{k}\right).\hat{n}\, ds = \iint_{S_1} \left((x^3 - yz)\hat{\imath} - 2x^2y\hat{\jmath} + z\hat{k}\right)\cdot \hat{\imath}\, dy\, dz$$

$$= \int_{z=0}^{2}\int_{y=0}^{2} (8 - yz)\ dy\, dz = \int_{z=0}^{2} \left[8y - \frac{zy^2}{2}\right]_0^2 dz = \int_{z=0}^{2} [16 - 2z]\ d$$

DEPARTMENT OF MATHEMATICS, RVCE

$$= \left[16z - \frac{2z^2}{2}\right]_0^2 = 32 - 4 = 28.$$

For S_2 $(OCEB)$ which is $xz - plane$, $y = 0$, $\hat{n} = -j$ & $ds = dz\,dx$.

$$\iint_{S_2} \left((x^3 - yz)\hat{\imath} - 2x^2y\hat{\jmath} + z\hat{k}\right).\hat{n}\,ds = \int_{x=0}^{2}\int_{y=0}^{2} (2x^2y)\ dz\,dx = 0 \qquad \because y = 0$$

For S_3 $(OADB)$ which is $xy - plane$, $z = 0$, $\hat{n} = -k$ & $ds = dx\,dy$.

$$\iint_{S_3} \left((x^3 - yz)\hat{\imath} - 2x^2y\hat{\jmath} + z\hat{k}\right).\hat{n}\,ds = \int_{y=0}^{2}\int_{x=0}^{2} (-z)\ dx\,dy = 0 \qquad \because z = 0$$

For S_4 $(OCFA)$ which is $yz - plane$, $x = 0$, $\hat{n} = -i$ & $ds = dy\,dz$.

$$\iint_{S_4} \left((x^3 - yz)\hat{\imath} - 2x^2y\hat{\jmath} + z\hat{k}\right).\hat{n}\,ds$$

$$= \int_{z=0}^{2}\int_{y=0}^{2} -(x^3 - yz)\ dy\,dz = \int_{z=0}^{2}\int_{y=0}^{2} yz\ dy\,dz = \int_{z=0}^{2} z\,dz \int_{y=0}^{2} y\,dy$$

$$= \left[\frac{z^2}{2}\right]_0^2 \left[\frac{y^2}{2}\right]_0^2 = 4.$$

For S_5 $(GFAD)$ which is parallel to $xz - plane$, *its equation is* $y = 2$, $\hat{n} = \hat{\jmath}$ & $ds = dx\,dz$.

$$\iint_{S_5} \left((x^3 - yz)\hat{\imath} - 2x^2y\hat{\jmath} + z\hat{k}\right).\hat{n}\,ds$$

$$= \int_{z=0}^{2}\int_{x=0}^{2} -2x^2y\ dx\,dz = \int_{z=0}^{2}\int_{x=0}^{2} -2x^2(2)\ dy\,dz = -4\int_{x=0}^{2} x^2\,dx \int_{z=0}^{2} 1\,dz =$$

$$= -4\left[\frac{x^3}{3}\right]_0^2 [z]_0^2 = -4 \times \frac{8}{3} \times 2 = -\frac{64}{3}.$$

For S_6 $(GECF)$ which is parallel to $xy - plane$, *its equation is* $z = 2$, $\hat{n} = k$ & $ds = dx\,dy$.

DEPARTMENT OF MATHEMATICS, RVCE

$$\iint_{S_6} \left((x^3 - yz)\hat{\imath} - 2x^2y\hat{\jmath} + z\hat{k}\right).\hat{n}\, ds$$

$$= \int_{x=0}^{2}\int_{y=0}^{2} z\ dy\, dx = 2\int_{x=0}^{2} 1\, dx \int_{y=0}^{2} 1\, dy = 8.$$

$$\therefore \iint_{S} \left((x^3 - yz)\hat{\imath} - 2x^2y\hat{\jmath} + z\hat{k}\right).\hat{n}\, ds = 28 + 0 + 0 + 4 - \frac{64}{3} + 8 = \frac{56}{3}.$$

Now to evaluate $\iiint_V \nabla.\vec{F}\, dV$

Consider

$$\nabla.\vec{F} = \left(\frac{\partial}{\partial x}\hat{\imath} + \frac{\partial}{\partial y}\hat{\jmath} + \frac{\partial}{\partial z}\hat{k}\right).\left((x^3 - yz)\hat{\imath} - 2x^2y\hat{\jmath} + z\hat{k}\right) = 3x^2 - 2x^2 + 1 = x^2 + 1$$

$$\therefore \iiint_V \nabla.\vec{F}\, dV = \int_{z=0}^{2}\int_{y=0}^{2}\int_{x=0}^{2} (x^2+1)dx\, dy\, dz = \int_{z=0}^{2} 1\, dz \int_{y=0}^{2} 1\, dy \int_{x=0}^{2} (x^2+1)\, dx$$

$$= [z]_0^2\, [y]_0^2 \left[\frac{x^3}{3} + x\right]_0^{2\,2} = (2)(2)\left(\frac{8}{3} + 2\right) = \frac{56}{3} \qquad \ldots\ldots\ldots\ldots(2)$$

From (1) and (2), we see that

$$\iint_S \vec{F}.\hat{n}\, ds = \iiint_V \nabla.\vec{F}\, dV = \frac{56}{3}$$

Hence the Gauss divergence theorem.

Problem 2. Verify divergence theorem $\vec{F} = xy\hat{\imath} - y\hat{\jmath} + 2z\hat{k}$ over the region bounded by the plane $x = 0, y = 0, z = 0$ & $2x + 2y + z = 4$.

Solution: Let $\emptyset = 2x + 2y + z - 4$

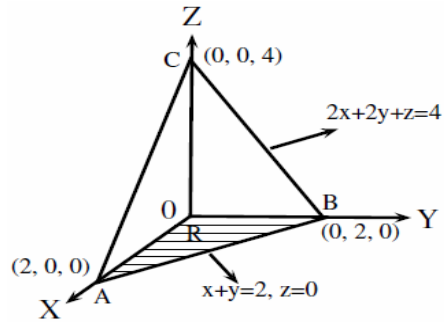
$$i.e. \quad \frac{x}{2} + \frac{y}{2} + \frac{z}{4} = 1$$

$$\hat{n} = \frac{\nabla\emptyset}{|\nabla\emptyset|} = \frac{2\hat{\imath} + 2\hat{\jmath} + \hat{k}}{3}$$

$$\vec{F}.\hat{n} = \frac{2xy - 2y + 2z}{3}$$

Now project the surface on $xy - plane$

DEPARTMENT OF MATHEMATICS, RVCE



$$|\hat{n}.\hat{k}| = \left|\frac{2\hat{\imath}+2\hat{\jmath}+\hat{k}}{3}.\hat{k}\right| = \frac{1}{3}$$

$x: 0 \; to \; 2$

$y: 0 \; to \; 2-x$

plane $2x+2y+z=4 \Rightarrow z = 4-2x-2y$

$$\therefore \iint_S \vec{F}.\hat{n}\, ds = \iint_R \vec{F}.\hat{n}\, \frac{dx\,dy}{|\hat{n}.\hat{k}|} = \iint_R \left(\frac{2xy-2y+2z}{3}\right) \frac{dx\,dy}{\frac{1}{3}}$$

$$= 2\int_{x=0}^{2}\int_{y=0}^{2-x} (xy-y+4-2x-2y)\, dy\, dx$$

$$= 2\int_{x=0}^{2}\int_{y=0}^{2-x} (xy-3y-2x-2y+4)\, dy\, dx$$

$$= 2\int_{x=0}^{2}\left[\frac{xy^2}{2}-\frac{3y^2}{2}-2xy+4y\right]_0^{2-x} dx$$

$$= 2\int_{x=0}^{2}\left[\frac{x(2-x)^2}{2}-\frac{3(2-x)^2}{2}-2x(2-x)+4(2-x)\right] dx$$

$$= 2\int_{x=0}^{2}\left[\frac{1}{2}(4x+x^3-4x^2)-\frac{3}{2}(4+x^2-4x)-8x+2x^2+8\right] dx$$

$$= 2\left[\frac{1}{2}\left(\frac{4x^2}{2}+\frac{x^4}{4}-\frac{4x^3}{3}\right)-\frac{3}{2}\left(4x+\frac{x^3}{3}-\frac{4x^2}{2}\right)-\frac{8x^2}{2}+\frac{2x^3}{3}+8x\right]_0^2 = 4 \qquad (1)$$

Surface	Remarks	$\hat{n}$	ds	$\vec{F}.\hat{n}$
$S_1: AOB$	$xy-plane$ $z=0$	$\hat{n} = -\hat{k}$	dx dy	-2z=0
$S_2: BOC$	$yz-plane$ $x=0$	$\hat{n} = -\hat{\imath}$	dy dz	-xy=0
$S_3: AOC$	$xz-plane$	$\hat{n} = -\hat{\jmath}$	dx dz	y=0

DEPARTMENT OF MATHEMATICS, RVCE

	$y=0$			
$S_4: ABC$	Projection on $xy-plane$	$\hat{n}=\frac{\nabla\emptyset}{\lvert\nabla\emptyset\rvert}$ $=\frac{2\hat{\imath}+2\hat{\jmath}+\hat{k}}{3}$	$\frac{dx\,dy}{\lvert\hat{n}.\hat{k}\rvert}$	$\vec{F}.\hat{n}=\frac{2(xy-y+4-2x-2y)}{3}$

Now consider

$$\nabla.\vec{F}=\left(\frac{\partial}{\partial x}\hat{\imath}+\frac{\partial}{\partial y}\hat{\jmath}+\frac{\partial}{\partial z}\hat{k}\right).\left(xy\hat{\imath}-y\hat{\jmath}+2z\hat{k}\right)=y+1$$

$$\therefore \iiint_V \nabla.\vec{F}\,dV=\int_{x=0}^{2}\int_{y=0}^{2-x}\int_{z=0}^{4-2x-2y}(y+1)\,dz\,dy\,dx$$

$$=\int_{x=0}^{2}\int_{y=0}^{2-x}[(y+1)z]_{z=0}^{4-2x-2y}dy\,dx$$

$$=\int_{x=0}^{2}\int_{y=0}^{2-x}[(y+1)(4-2x-2y)]dy\,dx$$

$$=\int_{x=0}^{2}\int_{y=0}^{2-x}[2y+4-2xy-2x-2y^2]dy\,dx$$

$$=\int_{0}^{2}\left[\frac{2y^2}{2}+4y-\frac{2xy^2}{2}-2xy-\frac{2y^3}{3}\right]_0^{2-x}dy$$

$$=\int_{0}^{2}\left[(2-x)^2+4(2-x)-x(2-x)^2-2x(2-x)-\frac{2}{3}(2-x)^3\right]dy=4 \quad \dots(2)$$

From (1) and (2), Gauss divergence theorem is verified.

Problem 3. Using divergence theorem, evaluate $\iint_S\left[(x^2-yz)\hat{\imath}+(y^2-zx)\hat{\jmath}+(z^2-xy)\hat{k}\right].\hat{n}\,ds$, over the surface of the rectangular parallelepiped $0\le x\le a, 0\le y\le b, 0\le z\le c$.

Solution: We have

$$\iint_S \vec{F}.\hat{n}\,ds=\iiint_V \nabla.\vec{F}\,dV$$

$$\nabla.\vec{F}=\left(\frac{\partial}{\partial x}\hat{\imath}+\frac{\partial}{\partial y}\hat{\jmath}+\frac{\partial}{\partial z}\hat{k}\right).\left((x^2-yz)\hat{\imath}+(y^2-zx)\hat{\jmath}+(z^2-xy)\hat{k}\right)=2(x+y+z)$$

$$\therefore \iiint_V \nabla.\vec{F}\,dV=\int_{z=0}^{c}\int_{y=0}^{b}\int_{x=0}^{a}2(x+y+z)\,dx\,dy\,dz$$

DEPARTMENT OF MATHEMATICS, RVCE

$$= 2\int_{z=0}^{c}\int_{y=0}^{b}\left(\frac{x^2}{2}+yx+zx\right)_0^a dy\,dz$$

$$= 2\int_0^c\int_0^b\left(\frac{a^2}{2}+ay+az\right)dy\,dz$$

$$= 2\int_0^c\left[\frac{a^2}{2}y+\frac{ay^2}{2}+azy\right]_0^b dz = 2\int_0^c\left[\frac{a^2b}{2}+\frac{ab^2}{2}+azb\right]dz$$

$$= \left[\frac{a^2b}{2}z+\frac{ab^2}{2}z+ab\frac{z^2}{2}\right]_0^c = abc(a+b+c).$$

Problem 4. Evaluate using divergence theorem $\iint_S \left[x^3\hat{\imath}+x^2y\hat{\jmath}+x^2z\hat{k}\right].\hat{n}\,ds$, where S is the surface consisting of the cylinder $x^2+y^2=a^2$ and the circular discs cut by the plane $z=0$ & $z=b$.

Solution: Here

$\vec{F} = x^3\hat{\imath}+x^2y\hat{\jmath}+x^2z\hat{k}$

$$\nabla.\vec{F} = \left(\frac{\partial}{\partial x}\hat{\imath}+\frac{\partial}{\partial y}\hat{\jmath}+\frac{\partial}{\partial z}\hat{k}\right).\left(x^3\hat{\imath}+x^2y\hat{\jmath}+x^2z\hat{k}\right) = 5x^2$$

$$\iint_S \vec{F}.\hat{n}\,ds = \iiint_V \nabla.\vec{F}\,dV = \iiint_V 5x^2dV$$

using cylindrical coordinates

$x = r\cos\theta,$

$y = r\sin\theta,$

$z = z,$

$dV = dx\,dy\,dz = r\,dr\,d\theta\,dz$

$r: 0\ to\ a$

$\theta: 0\ to\ 2\pi$

$z: 0\ to\ b$

$$\therefore \iiint_V 5x^2dV = \int_{z=0}^{b}\int_{\theta=0}^{2\pi}\int_{r=0}^{a} 5(r^2\cos\theta)\,r\,dr\,d\theta\,dz$$

$$= 5\int_{r=0}^{a} r^3dr\int_{\theta=0}^{2\pi}\cos^2\theta\,d\theta\int_{z=0}^{b}1\,dz = 5\frac{a^4}{4}\times\frac{1}{2}\times\left[\theta+\frac{\sin 2\theta}{2}\right]_0^{2\pi}\times[z]_0^b$$

$$= \frac{5a^4b\,\pi}{4}.$$

DEPARTMENT OF MATHEMATICS, RVCE

Problem 5. Using divergence theorem, $\iint_S \vec{F}.\hat{n}\, ds$, $\vec{F} = x^3\hat{\imath} + y^3\hat{\jmath} + z^3\hat{k}$ taken over the surface consisting of the hemisphere $x^2 + y^2 + z^2 = a^2$ above the $xy - plane$ bounded by the $xy - plane$.

Solution: Here

$\vec{F} = x^3\hat{\imath} + y^3\hat{\jmath} + z^3\hat{k}$

$$\nabla.\vec{F} = \left(\frac{\partial}{\partial x}\hat{\imath} + \frac{\partial}{\partial y}\hat{\jmath} + \frac{\partial}{\partial z}\hat{k}\right).\left(x^3\hat{\imath} + y^3\hat{\jmath} + z^3\hat{k}\right) = 3(x^2 + y^2 + z^2) = 3a^2$$

Using Spherical coordinates

$\because x = r\sin\theta\cos\phi\,, \qquad y = r\sin\theta\sin\phi\,, \qquad z = r\cos\theta$

$dx\, dy\, dz = r^2\sin\theta\ \ dr\, d\theta\, d\phi$

$r: 0\ to\ a$

$\theta: 0\ to \dfrac{\pi}{2} \qquad [verticle\ angle]$

$\phi: 0\ to\ 2\pi \qquad [Horizontal\ angle]$

$$\iint_S \vec{F}.\hat{n}\, ds = \iiint_V div(\vec{F})\ dV = 3\int_{\phi=0}^{2\pi}\int_{\theta=0}^{\frac{\pi}{2}}\int_{r=0}^{a} r^2(r^2\sin\theta)\, dr\, d\theta\, d\phi$$

$$= 3\int_{\phi=0}^{2\pi} 1\, d\phi \times \int_{\theta=0}^{\frac{\pi}{2}} \sin\theta\, d\theta \times \int_{r=0}^{a} r^4\, dr$$

$$= 3 \times 2\pi \times \frac{a^5}{5} \times 1 = \frac{6\pi a^5}{5}.$$

Using Cartesian coordinates

$$\iint_S \vec{F}.\hat{n}\, ds = \iiint_V div(\vec{F})\, dV$$

$$= \int_{x=-a}^{a}\int_{y=-\sqrt{a^2-x^2}}^{\sqrt{a^2-x^2}}\int_{z=0}^{\sqrt{a^2-x^2-y^2}} 3(x^2 + y^2 + z^2)\, dz\, dy\, dx = \frac{6\pi a^5}{5}$$

Exercise:

1. Verify divergence theorem for $\vec{F} = 4xz\hat{\imath} - y^2\hat{\jmath} + yz\,\hat{k}$ taken over the cube bounded by $x = 0, x = 1, y = 0, y = 1, z = 0, z = 1$.
2. Using divergence theorem, evaluate $\iint_S \vec{r}.\hat{n}\, ds$ where S is the surface of the sphere$x^2 + y^2 + z^2 = 9$. $Ans: 108\pi$

DEPARTMENT OF MATHEMATICS, RVCE

3. Using divergence theorem, evaluate $\iint_S \vec{F}.\hat{n}\, ds$ over the entire surface S of the region above xy plane bounded by the cone $x^2+y^2=z^2$ the plane $z=4$ where $\vec{F}=4xz\hat{\imath}-xyz^2\hat{\jmath}+3z\,\hat{k}$ $\quad Ans: 704\pi$

STOKES THEOREM

(Relation between line and surface integral)

Statement: If S be an open surface bounded by a simple closed curve C and $\vec{F}$ be any vector point function having continuous first order partial derivatives, then

$$\oint_C \vec{F}.d\vec{r} = \iint_S curl\,\vec{F}.\hat{n}\, ds = \iint_S curl\,\vec{F}.d\vec{s}$$

where $\hat{n}$ is the outward drawn unit normal at any point to S.

Problem 1. Verify Stokes theorem for $\vec{F}=(x^2+y^2)\hat{\imath}-2xy\hat{\jmath}$ taken round the rectangle bounded by the lines $x=\pm a, y=0, y=b$.

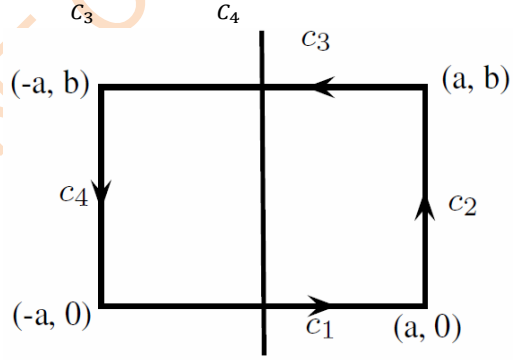
Solution: We have to prove that

$$\oint_C \vec{F}.d\vec{r} = \iint_S curl\,\vec{F}.\hat{n}\, ds$$

Now

$$\vec{F}.d\vec{r} = \left((x^2+y^2)\hat{\imath}-2xy\hat{\jmath}\right).\left(dx\,\hat{\imath}+dy\,\hat{\jmath}+dz\,\hat{k}\right) = (x^2+y^2)dx-2xy\,dy$$

$$\oint_C \vec{F}.d\vec{r} = \int_{C_1} \vec{F}.d\vec{r} + \int_{C_2} \vec{F}.d\vec{r} + \int_{C_3} \vec{F}.d\vec{r} + \int_{C_4} \vec{F}.d\vec{r} \qquad \dots\dots\dots\dots(1)$$



$$\oint_C \vec{F}.d\vec{r} = \oint_C (x^2+y^2)dx-2xy\,dy$$

Along $C_1: y=0 \Rightarrow dy=0,\ \ x: -a\ to\ a$

$$\therefore \int_{C_1} \vec{F}.d\vec{r} = \int_{C_1} (x^2+y^2)dx-2xy\,dy = \int_{-a}^{a} x^2dx = \frac{2a^3}{3}$$

DEPARTMENT OF MATHEMATICS, RVCE

Along $C_2: x = a \Rightarrow dx = 0,\ \ y: 0\ to\ b$

$$\therefore \int_{C_2} \vec{F}.d\vec{r} = \int_{C_2} (x^2 + y^2)dx - 2xy\, dy = \int 2ay\, dy = -ab^2$$

Along $C_3: y = b \Rightarrow dy = 0,\ \ x: a\ to - a$

$$\therefore \int_{C_3} \vec{F}.d\vec{r} = \int_{C_3} (x^2 + y^2)dx - 2xy\, dy = \int_{a}^{-a} (x^2 + b^2)\, dx = -\frac{2a^3}{3} - 2ab^2$$

Along $C_4: x = -a \Rightarrow dx = 0,\ \ y: b\ to\ 0$

$$\therefore \int_{C_4} \vec{F}.d\vec{r} = \int_{C_4} (x^2 + y^2)dx - 2xy\, dy = \int_{b}^{0} -2(-a)y\, dy = -ab^2$$

$$\therefore (1) \Rightarrow \oint_{C} \vec{F}.d\vec{r} = \frac{2a^3}{3} - ab^2 - \frac{2a^3}{3} - 2ab^2 - ab^2 = -4ab^2 \qquad \dots\dots\dots (2)$$

Next, consider

$$curl\, \vec{F} = \nabla \times \vec{F} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 + y^2 & -2xy & 0 \end{vmatrix} = -4y\hat{k}$$

Rectangle in $xy - plane \Rightarrow\ \hat{n} = \hat{k}\ and\ ds = dx\, dy$

$$\iint_S curl\, \vec{F}.\hat{n}\, ds\ = \iint_R -4y\hat{k}.\hat{k}\, dx\, dy = -\int_{y=0}^{b}\int_{x=-a}^{a} 4y\, dx\, dy = -4\int_{-a}^{a} 1\, dx \times \int_{0}^{b} y\, dy$$

$$\therefore \iint_S curl\, \vec{F}.\hat{n}\, ds\ = -4ab^2 \qquad \dots\dots\dots (3)$$

From (2) and (3), Stokes theorem is verified.

Problem 2. Verify Stokes theorem for $\vec{F} = (2x - y)\hat{\imath} - yz^2\hat{\jmath} - y^2z\hat{k}$ over the upper half surface of $x^2 + y^2 + z^2 = 1$ bounded by its projection on $xy - plane$.

Solution: The projection of upper half of the sphere $x^2 + y^2 + z^2 = 1$ in the $xy - plane$ $(z = 0)$ is the circle $x^2 + y^2 = 1$ and let C be its boundary.

We have

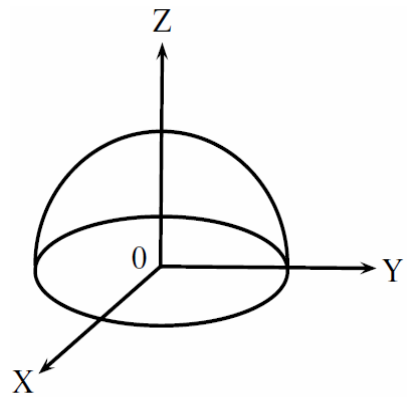
$$\oint_C \vec{F}.d\vec{r} = \iint_S (\nabla \times \vec{F}).\hat{n}\, ds$$

Consider

DEPARTMENT OF MATHEMATICS, RVCE

$$\oint_C \vec{F}.d\vec{r} = \int_C \{(2x-y)dx - yz^2dy - y^2z\,dz\}$$

In $xy-plane, z=0 \Rightarrow dz=0$



$$\therefore \oint_C \vec{F}.d\vec{r} = \int_C \{(2x-y)dx - yz^2dy - y^2z\,dz\} = \int_C (2x-y)dx$$

Here C is the circle $x^2+y^2=1$ whose parametric equation is given by

$x=\cos\theta, \qquad y=\sin\theta$

$\Rightarrow dx = -\sin\theta\, d\theta, \qquad dy = \cos\theta\, d\theta$

Here $\theta: 0\ to\ 2\pi$.

$$\therefore \oint_C \vec{F}.d\vec{r} = \int_0^{2\pi} (2\cos\theta - \sin\theta)(-\sin\theta)d\theta = \int_0^{2\pi} (\sin 2\theta + \sin^2\theta)\,d\theta = 0 + \int_0^{2\pi} \sin^2\theta\, d\theta$$

$$= 4\int_0^{\frac{\pi}{2}} \sin^2\theta\, d\theta = 4\times\frac{1}{2}\times\frac{\pi}{2} = \pi.$$

Next, consider

$$curl\,\vec{F} = \nabla\times\vec{F} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x-y & -yz^2 & -y^2z \end{vmatrix} = \hat{k}$$

On the xy −plane $\hat{n}=\hat{k}$ and $ds = dx\,dy$

$$\iint_S curl\,\vec{F}.\hat{n}\,ds = \iint_R \hat{k}\cdot\hat{k}\;dx\,dy = \iint_R 1\;dx\,dy$$

$= Area\ of\ circle\ (x^2+y^2=1) = \pi \qquad \because r=1$

Hence Stokes theorem is verified.

DEPARTMENT OF MATHEMATICS, RVCE

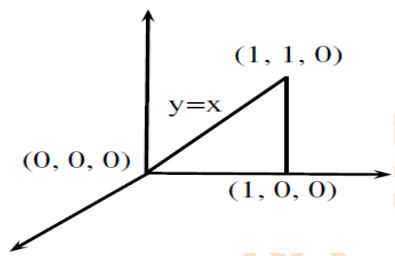
Problem 3. Evaluate by Stokes theorem $\oint_C (x+y)dx + (2x-z)dy + (y+z)dz$, C is the boundary of the triangular with vertices $(0,0,0), (1,0,0)$ and $(1,1,0)$.

Solution: By Stokes theorem we have

$$\oint_C \vec{F}.d\vec{r} = \iint_S curl\, \vec{F}.\hat{n}\, ds$$

$$\nabla \times \vec{F} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x+y & 2x-z & y+z \end{vmatrix} = 2\hat{\imath} + \hat{k}$$

In $xy - plane \Rightarrow \hat{n} = \hat{k}$ and $ds = dx\, dy$



$$\iint_S curl\, \vec{F}.\hat{n}\, ds = \iint_R (2\hat{\imath} + \hat{k}).\hat{k}\, dx\, dy = \iint_R 1\, dx\, dy = Area\ of\ the\ triangle$$

$$= \frac{1}{2} \times 1 \times 1 = \frac{1}{2}.$$

Exercise:

1. Evaluate $\oint_C xy\, dx + xy^2\, dy$ by Stoke's theorem where C is the square in the xy plane with vertices $(1,0)$ $(-1,0)$ $(0,1)$ $(0,-1)$.
2. Verify Stokes's theorem where $\vec{A} = (2x-y)\hat{\imath} - yz^2\hat{\jmath} - y^2z\,\hat{k}$ and S: upper half of the surface of the sphere $x^2 + y^2 + z^2 = 1$ $Ans: \pi$
3. Evaluate $\oint_C 4z\, dx - 2x\, dy + 2x\, dz$ by Stoke's theorem where C is the ellipse $x^2 + y^2 = 1$, $z = y + 1$. $Ans: -4\pi$

Video Links:

1. Line integral
https://www.youtube.com/watch?v=7FUNdFN6ZKI
2. Surface integral
https://www.youtube.com/watch?v=I1dfwKPV75A

Disclaimer: The content provided is prepared by department of Mathematics for the specified syllabus by using reference books mentioned in the syllabus. This material is specifically for the use of RVCE students and for education purpose only.

DEPARTMENT OF MATHEMATICS, RVCE